

NATIONAL CYBERSECURITY LAW

Design Project for VIT — Academic Assignment

Subject

Cybersecurity Law & Policy

ProjectType

Group / Individual Project

Total Marks

100Marks

DeliverablesReport+Presentation + Flowcharts +
Infographics**Important Notice to Students**

This is a comprehensive project assignment. Read all sections carefully before starting. Your team must draft a complete Cybersecurity Law for the fictional country of VIT's. Each part of this project corresponds to a section of the law you will create. Be creative, thorough, and technically accurate.

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PART 1: Introduction

Prepare an overview section that sets the foundation for your cybersecurity law. This section should clearly articulate why cybersecurity legislation is necessary for VIT's.

What to Include

- Purpose of the law — What is this law designed to achieve?
- Importance of cybersecurity — Why does VIT's need this law now?
- Challenges faced by VIT's — Describe specific threats from the scenario •
- Goals and expected outcomes — What will change after this law is enacted?

Scenario Context

VIT's is a rapidly growing digital nation with 50 million internet users, online banking, e-commerce platforms, government digital services, smart city infrastructure, AI-based public services, and cloud-based businesses. Recently, VIT's experienced large-scale data breaches, ransomware attacks on hospitals, banking fraud, social media misinformation campaigns, identity theft cases, and cyberbullying incidents.

PART 2: Definitions Section

Define at least 20 important cyber law terms. Your definitions must be precise, legally clear, and relevant to VIT's context. Below are the required terms — you may add more.

Required Terms to Define

	Term	Definition (To Be Completed by Student)
1	Cybercrime	Cybercrime refers to any illegal or unauthorized act carried out using a computer, network, or digital device as either a tool or a target. In the context of VIT's, this includes but is not limited to unauthorized access to systems, digital fraud, online harassment, data theft, and the deployment of malicious software against individuals, organizations, or government infrastructure.
2	Hacker	A hacker is any individual who attempts to gain unauthorized access to computer systems, networks, or digital devices by exploiting technical vulnerabilities, software flaws, or weak security configurations. The term broadly encompasses individuals with varying levels of intent — from those with malicious purposes to security researchers testing system resilience.
3	Ethical Hacker	An ethical hacker, also known as a white-hat hacker or penetration tester, is a cybersecurity professional who is expressly authorized by an organization to probe its systems, networks, and applications for security weaknesses. Their goal is to identify and report vulnerabilities before malicious actors can exploit them, thereby strengthening the overall security posture of the organization.
4	Malware	Malware, short for malicious software, refers to any software program or code intentionally designed to damage, disrupt, or gain unauthorized access to a computer system or network. This category encompasses viruses, worms, Trojans, spyware, adware, and ransomware, all of which can compromise data integrity, confidentiality, and system availability.
5	Ransomware	Ransomware is a type of malicious software that encrypts or locks a victim's files, systems, or data and demands a ransom payment — typically in cryptocurrency — in exchange for the decryption key or restoration of access. Ransomware attacks have targeted hospitals, financial institutions, and government services in VIT's, causing significant operational and financial damage.
6	Phishing	Phishing is a form of social engineering in which an attacker impersonates a legitimate and trusted entity — such as a bank, government agency, or well-known service provider — to deceive individuals into disclosing sensitive information such as login credentials, credit card numbers, or personal identification details. Attacks are most commonly carried out via deceptive emails, fake websites, or fraudulent messages.
7	Data Breach	A data breach is a security incident in which unauthorized individuals gain access to confidential, protected, or sensitive data without the permission of the data owner or custodian. Breaches may result from external cyberattacks, insider threats, or accidental exposure. In VIT's, organizations are legally obligated to report confirmed data breaches to the NCSA within 72 hours of detection.
8	Personal Data	Personal data refers to any information that identifies or can reasonably be used to identify a specific living individual. This includes direct identifiers such as full name, national identification number, date of birth, home address, email address, and phone number, as well as indirect identifiers such as IP addresses, device identifiers, and location data that, when combined, can identify a person.
9	Sensitive Data	Sensitive data is a subset of personal data that carries a higher risk of harm if improperly disclosed or misused. In VIT's legal framework, sensitive data includes information relating to a person's health and medical records, biometric data, genetic information, financial account details, religious or political beliefs, racial or ethnic origin, sexual orientation, and any data relating to minors.
10	Digital Signature	A digital signature is a cryptographic mechanism used to verify the authenticity and integrity of a digital message, document, or transaction. It is the digital equivalent of a handwritten signature, providing assurance that the document was created by a known sender and has not been altered in transit. In VIT's, digital signatures issued by accredited Certifying Authorities hold legal validity equivalent to physical signatures.
11	Electronic Evidence	Electronic evidence refers to any data, records, or information stored or transmitted in digital form that may be used in a legal proceeding to prove or disprove a fact. This includes emails, chat logs, system logs, digital images, video recordings, transaction records, metadata, and data recovered from storage devices. Electronic evidence collected in VIT's must follow prescribed forensic procedures to ensure admissibility in court.
12	Identity Theft	Identity theft is the deliberate and unauthorized acquisition, use, or misuse of another person's personal identifying information — such as their name, national ID number, financial account details, or biometric data — for fraudulent purposes. Common goals include financial gain, accessing services, or evading law enforcement. Identity theft is treated as a serious criminal offense under VIT's Cybersecurity Law.
13	Artificial Intelligence	Artificial Intelligence (AI) refers to the simulation of human cognitive functions — such as learning, reasoning, problem-solving, and language understanding — by computer systems and algorithms. In the context of VIT's legal framework, AI systems used in critical infrastructure, public decision-making, healthcare, and law enforcement are subject to mandatory transparency, accountability, and security compliance standards.
14	Cloud Computing	Cloud computing is a model of delivering computing services — including servers, storage, databases, networking, software, and analytics — over the internet on a pay-as-you-use basis. Organizations in VIT's that store or process personal or government data using cloud infrastructure are required to comply with data residency requirements, ensure encryption at rest and in transit, and conduct regular security audits of their cloud environments.
15	Critical Infrastructure	Critical infrastructure refers to the systems, services, assets, and networks whose disruption, compromise, or destruction would have a significant adverse impact on national security, public health, economic stability, or the safety of VIT's citizens. Designated critical infrastructure sectors in VIT's include banking and finance, healthcare, energy and power generation, transportation, telecommunications, water systems, and government data centers.
16	Cyberbullying	Cyberbullying is the use of digital technologies — including social media platforms, messaging applications, online forums, and gaming networks — to harass, threaten, intimidate, humiliate, or cause emotional harm to another individual. In VIT's, cyberbullying directed at minors is treated with particular severity, and platforms are required to provide accessible reporting mechanisms and respond to verified complaints within 24 hours.
17	Deepfake	A deepfake is a synthetic media artifact — typically a video, image, or audio recording — generated using artificial intelligence techniques such as deep learning and generative adversarial networks (GANs) to realistically portray individuals saying or doing things they never actually said or did. In VIT's, the creation or distribution of deepfakes for defamatory, non-consensual intimate, or politically manipulative purposes constitutes a criminal offense.
18	Encryption	Encryption is the process of converting readable plaintext data into an unreadable format — known as ciphertext — using a cryptographic algorithm and a key, such that only authorized parties with the corresponding decryption key can restore the original data. Under VIT's Cybersecurity Law, all organizations handling personal or sensitive data are mandated to implement strong, industry-standard encryption protocols for data at rest and in transit.
19	Firewall	Encryption is the process of converting readable plaintext data into an unreadable format — known as ciphertext — using a cryptographic algorithm and a key, such that only authorized parties with the corresponding decryption key can restore the original data. Under VIT's Cybersecurity Law, all organizations handling personal or sensitive data are mandated to implement strong, industry-standard encryption protocols for data at rest and in transit.

20	Zero-Day Exploit	A zero-day exploit is a cyberattack that takes advantage of a previously unknown software vulnerability for which no official patch or fix has yet been developed or released by the vendor. Because defenders have zero days to prepare, such exploits are particularly dangerous. Organizations in VIT's that discover zero-day vulnerabilities in widely-used systems are legally required to report them to the NCSA and follow responsible disclosure guidelines.
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PART 3: Citizens' Digital Rights

Create a comprehensive Bill of Digital Rights for every citizen of VIT's. For each right, provide a clear explanation of what it means and how it will be enforced.

Required Rights

Right to Privacy

Every citizen of VIT's has the right to privacy and protection of their personal information. No organization or government agency may collect, store, process, or share personal data without a lawful reason and the citizen's consent. Any unauthorized access, misuse, or disclosure of personal information shall be considered a violation of this right and may result in legal penalties. Citizens may file complaints with the NCSA, which shall investigate and take enforcement action against offenders.

Right to Secure Data

Every citizen has the right to expect that organizations will protect their personal and sensitive data using industry-standard security measures. Organizations must implement encryption, access controls, security monitoring, and regular security audits to safeguard data from cyber threats. Any organization that fails to adequately protect citizen data and causes a breach may face financial penalties and compensation claims.

Right to Data Deletion

Every citizen has the right to request the deletion of their personal data when it is no longer required for legitimate purposes. Organizations must verify the request and permanently remove the data from their systems within 30 days unless retention is required by law. Citizens may appeal to the NCSA if an organization refuses to delete eligible data without a valid reason.

Right to Information

Every citizen has the right to know what personal data is being collected about them, why it is being collected, how it is being used, and with whom it is being shared. Organizations must provide clear privacy notices and respond to citizen requests regarding their personal data within 30 days. Failure to provide accurate information may result in fines and regulatory action by the NCSA.

Right to Report Cybercrime

Every citizen has the right to report cybercrimes, cyberbullying, online fraud, identity theft, and other cyber incidents without fear of retaliation or discrimination. Reports may be submitted through cybercrime portals, police stations, or directly to the NCSA. Authorities must acknowledge complaints within 24 hours and initiate appropriate action within the legally prescribed time limits to ensure prompt justice and protection for victims.

Right to Fair Investigation

Any individual accused of a cybercrime has the right to a fair, transparent, and timely investigation. Law enforcement agencies and the NCSA must collect evidence legally, respect due process, and ensure that no person is punished without sufficient proof. Citizens also have the right to legal representation and access to information regarding the progress of their case.

Instruction

You must include at least 6 rights. You may add more. Each right should be at least 2-3 paragraphs with specific implementation details

PART 4: Cyber Offenses and Crimes

Define categories of cybercrime. For each offense, specify a description, real-world example within VIT's context, criminal punishment, and financial fine. Complete the table below.

Crime	Description + Example	Punishment	Fine
Unauthorized Access	Description: Sneaking into a computer network, server, or digital system without explicit permission. VIT Example: A student hacks into a professor's office desktop to look at upcoming quiz questions before they are published.	Up to 3 years imprisonment	Civil liability — compensation to the affected party
Data Theft	Description: Copying, downloading, or stealing private digital assets, proprietary source code, or databases. VIT Example: A student worker in the printing room copies the final mathematics exam paper onto a hidden USB drive to distribute it.	Up to 3 years imprisonment	Up to ₹10 Lakhs
Identity Theft	Description: Stealing someone else's digital credentials, login keys, or personal identifiers to pretend to be them. VIT Example: A student uses a hardware keylogger to capture a senior professor's login details and access the grading network.	Up to 3 years imprisonment	Up to ₹5 Lakhs
Phishing	Description: Sending deceptive emails or links designed to look legitimate to trick targets into handing over passwords or money. VIT Example: Attackers blast a fake "Exam Timetable PDF" link to campus emails to capture student portal passwords.	Up to 3 years imprisonment	Up to ₹7 Lakhs
Ransomware	Description: Deploying malicious software that encrypts data and holds a system hostage until a ransom is paid. VIT Example: Hackers lock the central fee registry during registration week and demand ₹25,000 per student to unlock it.	Up to 10 years imprisonment	Up to ₹25 - ₹50 Lakhs
Malware Distribution	Description: Intentionally creating and spreading viruses, spyware, or malicious payloads to disrupt computers. VIT Example: Injecting a malicious script into the campus open Wi-Fi network that automatically infects connected student phones.	Up to 3 years imprisonment	Up to ₹5 Lakhs
Cyberbullying	Description: Using digital channels to systematically attack, humiliate, intimidate, or terrify another individual. VIT Example: Trolls anonymously flood an underperforming student's social accounts with malicious, threatening messages.	Up to 7 years imprisonment	Up to ₹5 Lakhs
Online Harassment	Description: Persistent, unwelcome digital tracking, messaging, or communication that causes severe mental distress. VIT Example: Creating a fake, malicious profile using a peer's name to send insulting messages across student chat groups.	Up to 5 years imprisonment	Civil liability — compensation to the affected party
Financial Fraud	Description: Using computer resources to execute deceptive transactions, drain accounts, or steal money illegally. VIT Example: Setting up a lookalike university payment gateway to intercept and steal student tuition fee transfers.	Up to 7 years imprisonment	Up to ₹25 - ₹50 Lakhs
Misinformation	Description: Deliberately spreading unverified, harmful, or fabricated news online to stir up public panic or chaos. VIT Example: Circulating a fabricated notice on WhatsApp claiming all university exams are canceled due to a fictitious emergency.	Up to 3 years imprisonment	Up to ₹5 Lakhs
Website Defacement	Description: Unauthorized hacking and altering of a website's public visual interface to leave a malicious message. VIT Example: Changing the main college homepage screen to show a mocking message from a hostile hacker collective.	Up to 5 years imprisonment	Up to ₹5 Lakhs
Cyber Terrorism	Description: Actively attacking infrastructure or public networks to threaten national safety, utilities, or critical services. VIT Example: Hacking the smart-grid solar power relays to intentionally trigger a complete blackout during final exams.	Up to 5 years imprisonment	Civil liability — compensation to the affected administration

PART 5: Data Protection Framework

Create a comprehensive data protection framework that all organizations operating in VIT's must follow. Your framework must cover the full data lifecycle.

Data Collection Rules

- Collect only necessary data.
- Inform users before collecting data.
- Obtain user consent.
- Do not collect sensitive data without permission.

Data Storage Rules

- Encrypt all sensitive information.
- Store data on secure servers.
- Restrict access to authorized employees only.
- Perform regular backups.

Data Sharing Rules

- Share data only with user consent.
- Use secure channels for data transfer.
- Maintain records of all data sharing activities.

User Consent Requirements

- Consent must be clear and voluntary.
- Users can withdraw consent anytime.
- Separate consent must be taken for sensitive data.

Data Retention Period

- Personal data should be retained only as long as required.
- Unused data must be deleted after 5 years unless legally required.

Data Deletion Process

- *Users may request deletion.*
- *Organizations must verify identity.*
- *Data must be permanently removed within 30 days.*

Breach Notification Requirements

- Report breaches within 72 hours.
- Inform affected users immediately.
- Provide details of affected data and corrective actions.

PART 6: Corporate Responsibilities

All companies and organizations operating in VIT's must comply with the following cybersecurity responsibilities. Create at least 10 specific, enforceable corporate obligations.

Required Responsibilities (Minimum 10)

Responsibility 1: [Title]

- What it means: Companies cannot just rush software out the door and patch it later. Development teams must build security checks directly into their daily coding workflows. Every time a developer writes code or updates an app, automated tools must scan the code and its third-party building blocks for security bugs, logic flaws, and outdated libraries before that software ever reaches a customer.
- Who this applies to: Any business inside VIT's that writes its own software, builds mobile apps, runs e-commerce portals, or develops financial technology.
- The cost of ignoring it: A corporate fine of ₹12.50 Crore. Additionally, the government will halt all public software updates until an independent security audit proves the development pipeline is clean and secure.

Responsibility 2: [Locking Data Down with Encryption]

- What it means: Companies have an absolute duty to protect customer records from prying eyes. All personal and financial data must be scrambled using industry-standard encryption, making it unreadable both at rest (while stored on databases and hard drives) and in transit (while traveling over the internet via secure HTTPS/TLS web tunnels).
- Who this applies to: Online banking services, shopping portals, medical records systems, and public digital applications.
- The cost of ignoring it: A massive flat fine of ₹41.75 Crore, and the company will be legally ordered to stop collecting or processing any data until their cryptographic setups are fixed.

Responsibility 3: [Mandating Multi-Factor Authentication (MFA)]

- What it means: You should not wait for real hackers to find the holes in your defense. Every year, companies must hire independent, certified third-party ethical hackers to aggressively attack their networks and systems. This formal vulnerability assessment and penetration testing (VAPT) maps out security gaps so they can be patched before they are exploited.
- Who this applies to: All cloud service operators, fintech platforms, smart city networks, and critical national infrastructure systems.
- The cost of ignoring it: A fine of ₹20.87 Crore. The business will also be placed on a "critical watch list," triggering mandatory, unannounced government security audits.

Responsibility 4: [Building Strong Network Borders (DMZ Perimeters)]

- What it means: Companies cannot use "flat" networks where a breach on a basic employee laptop gives a hacker immediate access to private consumer records. Networks must be separated into secure compartments using modern firewalls. Public-facing web servers must sit in an isolated outer zone—a Demilitarized Zone (DMZ)—making it physically impossible for external traffic to talk directly to core back-end identity databases.
- Who this applies to: Online cloud vendors, e-commerce networks, and public web service operations.
- The cost of ignoring it: The company's public-facing websites and apps will be shut down by the government until network engineers rebuild compliant, separated network borders.

Responsibility 5: [Taking Blame for Third-Party Vendors]

- What it means: You cannot pass the buck when it comes to security. Before a company connects its systems to any outside software partner, external API, or outsourced cloud tool, it must perform deep security vetting. Businesses are legally responsible for the security standards of the vendors they choose to do business with.
- Who this applies to: Any business utilizing third-party payment infrastructure, outsourced developer code, or SaaS integrations.
- The cost of ignoring it: The primary firm takes full legal co-liability for any data breaches caused by a vendor's broken code, alongside a corporate fine of ₹62.62 Crore.

Responsibility 6: [System Logs Data Breach]

- What it means: When a major security incident occurs, the first thing attackers do is wipe the system logs to cover their tracks. To stop this, companies must stream all system records—such as database queries, configuration changes, and admin logins—to an isolated, append-only log server in real time. These records must be digitally locked so that no one, not even a high-level system administrator, can change or delete them.
- Who this applies to: Cloud infrastructure vendors, data centers, banking platforms, and critical utility control systems.
- The cost of ignoring it: A structural fine of ₹25.00 Crore for failing to protect audit records. Furthermore, if investigators prove that a company intentionally wiped its logs to hide an active data breach, corporate officers will face criminal charges and prison time under Section 66.

Responsibility 7: [Locking Down Physical Server Rooms]

- What it means: Cybersecurity is useless if someone can walk into a building and plug a malicious device directly into a server. Data centers and technical closets must be treated like high-security vaults. Access points must require a combination of biometric data (like fingerprints or facial scans) and physical ID badges, backed by continuous 24/7 CCTV surveillance and two years of archived entry logs.
- Who this applies to: Server colocation centers, corporate data rooms, private hospitals, and transport dispatch systems.
- The cost of ignoring it: A flat ₹10.00 Crore fine per unmonitored server zone, along with an immediate shutdown of the facility until physical access controls are verified.

Responsibility 8: [Securing Remote Work and BYOD (Bring Your Own Device) Policies]

- What it means: When employees work from home or use personal phones for office work, they open up new entry points for hackers. Companies must enforce strict Mobile Device Management (MDM) rules. Any personal device used for work must run a secure, company-monitored profile that isolates work data, blocks unauthorized app installations, and allows the IT team to wipe all company data remotely if the device is lost or stolen.
- Who this applies to: Any organization inside VIT's that permits hybrid work schedules, remote access, or the use of personal devices for corporate tasks.
- The cost of ignoring it: A corporate fine of ₹7.50 Crore and an immediate suspension of all remote connection privileges until an MDM system is deployed.

Responsibility 9: [Mandatory Implementation of DNS Security Extension (DNSSEC)]

- What it means: Hackers love to hijack web traffic by altering domain records, pointing users to a cloned website that looks exactly like the real thing to steal their logins. Public websites must use DNSSEC (Domain Name System Security Extensions). This adds a digital signature to a company's web address records, confirming to a user's browser that the site they are visiting is legitimate and has not been maliciously redirected.
- Who this applies to: All public banking portals, e-commerce applications, government registries, and public web service utilities.
- The cost of ignoring it: Temporary suspension of the company's domain hosting privileges within VIT's registry and an administrative fine of ₹3.00 Crore.

Responsibility 10: [Secure API (Application Programming Interface) Perimeters]

- What it means: Modern web apps talk to backend databases via APIs, making APIs a favorite target for attackers looking to extract information in bulk. Companies must secure all API endpoints using strict authorization checks, automated rate-limiting to prevent automated data scraping, and constant traffic monitoring to flag and drop malformed injection attempts before they process on backend services.
- Who this applies to: Any business utilizing microservice architectures, cloud-integrated applications, third-party plugin extensions, or online service tools.
- The cost of ignoring it: A corporate fine of ₹22.00 Crore for negligent application endpoint design, paired with immediate containment orders to deactivate unmonitored or open data pathways.

Examples to Consider

Maintain security controls | Conduct regular security audits | Train employees on cybersecurity
| Report incidents within 72 hours | Protect customer information | Appoint a Data Protection
Officer | Conduct annual penetration testing | Implement access control policies | Encrypt
sensitive data | Maintain incident response plans

PART 7: Government Responsibilities

Specify how government agencies of VIT's must protect national interests in cyberspace. Include concrete mechanisms for each area.

Protect National Infrastructure

The Government shall identify banking systems, healthcare facilities, airports, railways, power plants, telecommunications networks, and government data centers as critical national infrastructure. These sectors must implement strong cybersecurity measures including encryption, multi-factor authentication, network segmentation, continuous monitoring, regular security audits, and disaster recovery plans. The NCSA shall conduct periodic inspections to ensure compliance and maintain national security.

Monitor Cyber Threats

The Government shall establish a 24×7 Cyber Threat Monitoring Center under the NCSA to continuously monitor national networks and critical infrastructure. The center will collect threat intelligence, analyze suspicious activities, detect emerging cyber threats, and issue real-time security alerts to organizations and government agencies. Advanced monitoring systems, intrusion detection tools, and threat intelligence platforms shall be used to strengthen national cyber defense capabilities.

Investigate Cybercrimes

The Government shall investigate cybercrimes through specialized cybercrime units and the NCSA Digital Forensics Laboratory. Investigators shall collect digital evidence, perform forensic analysis, obtain legal warrants when necessary, and coordinate with law enforcement agencies to identify offenders. All investigations shall be conducted fairly, transparently, and in accordance with established legal procedures to ensure justice and protect citizens' rights.

International Coordination

VIT's shall cooperate with INTERPOL, foreign governments, and international cybersecurity organizations to combat cybercrime and cyber threats that cross national borders. The Government shall participate in intelligence-sharing programs, joint investigations, cyber defense exercises, and international cybersecurity agreements to improve global cooperation and strengthen national cybersecurity preparedness.

Awareness Programs

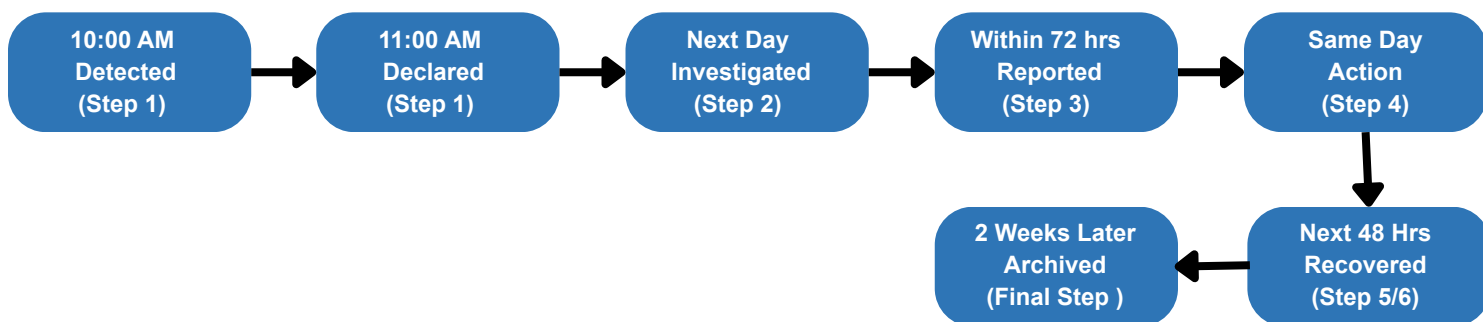
The Government of VIT's shall conduct nationwide cybersecurity awareness campaigns through schools, colleges, workplaces, television, radio, and social media platforms. These programs will educate citizens about safe internet usage, phishing scams, cyberbullying, online fraud, password security, and data privacy. The NCSA shall organize annual Cyber Safety Weeks, public workshops, and awareness drives to ensure that citizens of all age groups can identify and respond to cyber threats effectively.

PART 8: Cyber Incident Reporting Framework

Design a step-by-step process for how cyber incidents are reported, investigated, and resolved in VIT's. You must also create a flowchart (as a separate deliverable).

Reporting Process Steps

	Stage	Description (Student to Complete)
1	Incident Detected	During morning registration week, computer lab screens across the campus suddenly freeze and flash a fake message telling students to pay a ₹25,000 fee via a crypto link. The administration realizes the entire system is locked, meaning they cannot try to handle it quietly or hide it from records. The IT desk immediately logs the event, takes a snapshot of the corrupted memory, and triggers the campus emergency mode.
2	Internal Investigation	The campus tech team scrambles to figure out how the hackers managed to sneak past their network defenses. They discover a student opened a fake "exam timetable" email while connected to the open campus Wi-Fi, allowing the virus to spread straight to the administrative servers. The team completes its investigation within 24 hours, proving the fee logs are locked but the main grading records are safe.
3	Report Authority to	With registration lines growing and staff forced to process thousands of students using slow, manual paperwork, the college knows it must notify the government. Within the mandatory 72-hour legal window, the university's data officer logs onto the national portal to report the breach. The report lists the specific ransomware strain, the email footprint, and the 10,000 affected student accounts.
4	Risk Assessment	Because this attack targets a major university during its busiest week and compromises student financial profiles, the case gets flagged immediately. The campus security head meets with the government's threat team to evaluate the operational damage and data privacy risks. Following this joint review, the authority officially labels the incident as a High Severity Threat to national interests.
5	Mitigation	To stop the malware from crawling into payroll, faculty grading portals, or the library network, the IT team goes into isolation mode. Following the government's technical orders, they shut down the campus Wi-Fi and block the attacker's external IP addresses. They force a global password reset for all staff and apply urgent security patches to the remaining clean systems.
6	Recovery	With the network borders secure and the hackers locked out, the university refuses the ransom demands and focuses on rebuilding. Technicians completely format the corrupted hard drives, reflash system firmware, and begin pulling records from their secure, offline backups. The team sends daily progress updates to the government until the registration portal is safely brought back online.
7	Final Report	Two weeks later, with student systems running smoothly and fees processing normally, the college completes its final legal requirement. The security head writes a retrospective report detailing the root cause, financial losses, and precise containment timelines. To officially close the case file with the NCSA, the report includes a signed promise showing they have upgraded campus email filters and enabled mandatory Multi-Factor Authentication (MFA).



Deliverable Reminder

You must submit a separate FLOWCHART showing this complete process. Use tools like Canva, draw.io, Lucidchart, or PowerPoint SmartArt to create a professional diagram.

PART 9: National Cyber Security Authority (NCSA)

Design the organizational structure, responsibilities, and powers of the NCSA — the primary government agency responsible for cybersecurity in VIT's.

Organizational Structure

The National Cyber Security Authority (NCSA) shall be the apex cybersecurity agency of VIT's, functioning under the Ministry of Digital Security. It shall be headed by a Director General assisted by Deputy Directors responsible for operations, investigations, policy, and technology. The organization will maintain regional cyber offices across the country to ensure effective coordination, rapid incident response, and implementation of cybersecurity policies. All departments will report to the Director General through a defined chain of command to ensure accountability and efficient management.

Core Responsibilities

The NCSA shall be responsible for developing national cybersecurity policies, protecting critical infrastructure, investigating cybercrimes, coordinating responses to cyber incidents, conducting security audits, promoting cybersecurity awareness, and strengthening the nation's cyber resilience. It shall also issue threat advisories, maintain national cybersecurity standards, support research and innovation, and collaborate with public and private organizations to ensure a secure digital environment for all citizens of VIT's.

Legal Powers

The NCSA shall have the legal authority to conduct cybersecurity inspections, order security audits, request information and digital evidence from organizations, impose penalties for non-compliance, and direct emergency cybersecurity measures during major incidents. In cases involving serious threats to national security, the authority may temporarily block malicious websites, suspend vulnerable services, coordinate with law enforcement agencies, and recommend criminal prosecution of individuals or organizations responsible for cyber offenses.

Investigation Process

When a cybercrime complaint is received, the NCSA shall first conduct a preliminary assessment to determine its severity and legitimacy. Investigators will then collect digital evidence, perform forensic analysis, identify the source and impact of the attack, and coordinate with relevant organizations and law enforcement agencies. After completing the investigation, a detailed report containing findings, evidence, recommendations, and legal actions will be prepared, and measures will be implemented to prevent similar incidents from occurring in the future.

Departments

1. Threat Intelligence Unit (TIU)
 - Monitor cyber threats 24/7.
 - Track hackers and cybercriminal groups.
 - Share threat alerts with organizations.
 - Maintain national threat databases.
2. Incident Response Team (IRT)
 - Respond to cyber emergencies.
 - Coordinate recovery after cyberattacks.
 - Provide technical assistance to affected organizations.
 - Manage national cyber crisis situations.
3. Digital Forensics Laboratory
 - Analyze digital evidence.
 - Recover deleted files and logs.
 - Support police investigations.
 - Prepare forensic reports for courts.
4. Data Protection Division
 - Ensure compliance with privacy laws.
 - Conduct data protection audits.
 - Handle citizen complaints regarding misuse of personal data.
 - Recommend penalties for violations.
5. Public Awareness and Education Division
 - Conduct cybersecurity awareness campaigns.
 - Create educational materials.
 - Organize workshops and seminars.
 - Promote safe internet usage.

PART 10: AI and Emerging Technology Regulations

Create specific cybersecurity regulations for emerging technologies used in VIT's. For each technology, identify key risks and write enforceable rules.

Artificial Intelligence

Key Risks

Hackers "poisoning" training datasets to manipulate an AI's automated choices, or using automated AI scripts to hunt for open campus network ports and write custom malware code in seconds.

Regulations

Mandatory anti-tampering validation checks for deployed algorithms, transparent audit logs for training datasets, and strict human-in-the-loop overrides for sensitive data systems.

Deepfakes

Key Risks

Hyper-realistic voice and video clones of university deans, directors, or officials used in social engineering scams to trick campus accounts desks into wiring emergency funds.

Regulations

Mandatory cryptographic watermarking embedded by all synthetic media tools; creating or sharing unauthorized fakes to commit fraud carries immediate criminal charges.

IoT Devices

Key Risks

Cheap campus gadgets (smart routers, bio-locks) with weak processing power being hijacked into massive botnets to launch devastating volumetric DDoS attacks.

Regulations

Unchangeable default admin passwords are strictly banned; all connected hardware must support secure wireless firmware patches or face a mandatory product recall.

Digital Identity Systems

Key Risks

Centralized biometric registries becoming high-value targets for state-sponsored identity theft, unauthorized user tracking, and credential-stuffing loops.

Regulations

Identities must rely on decentralized architectures and robust asymmetric encryption; biometric verification checks must process locally on edge hardware with open audit logs for users

Autonomous Systems

Key Risks

Attackers hacking wireless command links on automated transport shuttles or drone routes, or tampering with smart solar power relays to trigger massive blackouts.

Regulations

Control networks must run on segmented subnets completely cut off from the public internet, featuring hardwired, physical manual overrides for immediate human safety intervention.

Cloud Computing

Key Risks

A single misconfigured access control rule exposing massive blocks of student data to the public internet, or data leakage on shared, multi-tenant servers.

Regulations

Providers must use absolute logical data isolation on shared hardware; cloud storage must default to private and store citizen data locally within the nation's borders.

PART 11: Critical Infrastructure Protection

Create sector-specific cybersecurity requirements for each of the following critical infrastructure types in VIT's. Requirements should be practical and technically grounded.

Sector	Cybersecurity Requirements (Student to Complete)
Banks & Financial	Banks must use Multi-Factor Authentication (MFA), encrypt customer data, monitor fraudulent transactions, conduct regular security audits, and report major cyber incidents to the NCSA within 72 hours
Hospitals & Healthcare	Hospitals must encrypt patient records, restrict access to authorized staff, maintain secure backups, regularly update medical systems, and establish emergency response plans against cyberattacks.
Airports & Aviation	Airports must secure air traffic systems, implement strict access controls, continuously monitor networks, conduct security testing, and maintain backup communication systems for emergencies.
Railways & Transport	Railway systems must use strong authentication, encrypt passenger data, monitor networks for threats, conduct regular cybersecurity drills, and maintain disaster recovery plans.
Power Plants & Energy	Power plants must isolate control systems from the internet, monitor operational networks, apply security updates regularly, conduct risk assessments, and maintain disaster recovery plans
Government Data Centers	Government data centers must encrypt sensitive information, enforce Multi-Factor Authentication, maintain security logs, conduct regular audits, and store secure backups at separate locations.

PART 12: Awareness and Education

Design a National Cybersecurity Awareness Campaign for VIT's. Your campaign must reach all segments of society through age-appropriate and effective programs.

School Programs (Age 6-18)

- *Curriculum Topics:* Digital citizenship basics, safe password creation, identifying online grooming behaviors, and the fundamentals of cyberbullying prevention. As students enter high school, topics expand to safe social media use, recognizing phishing attempts, and understanding the legal consequences of digital harassment.
- *Activities:* Interactive classroom gamification, mock-phishing email identification challenges, and annual poster-making contests focused on digital safety themes. Schools will host a mandatory "Cyber Safety Week" featuring guest lectures from ethical hacking professionals.
- *Learning Outcomes:* Students will develop a strong baseline understanding of digital footprints, know exactly how to report online harassment safely, and recognize malicious download links to protect personal data.

College & University Programs

- *Curriculum Topics:* Digital citizenship basics, safe password creation, identifying online grooming behaviors, and the fundamentals of cyberbullying prevention. As students enter high school, topics expand to safe social media use, recognizing phishing attempts, and understanding the legal consequences of digital harassment.
- *Activities:* Interactive classroom gamification, mock-phishing email identification challenges, and annual poster-making contests focused on digital safety themes. Schools will host a mandatory "Cyber Safety Week" featuring guest lectures from ethical hacking professionals.
- *Learning Outcomes:* Students will develop a strong baseline understanding of digital footprints, know exactly how to report online harassment safely, and recognize malicious download links to protect personal data.

Corporate Training Programs

- *Mandatory Requirements:* Every registered business must provide a formal cybersecurity onboarding brief to new hires, combined with mandatory role-based access security training for handling sensitive personal customer data.
- *Frequency:* Security training sessions must be executed every 6 months across all corporate divisions to address evolving social engineering techniques.
- *Assessment Methods:* Unannounced internal phishing simulations deployed by the IT department to measure employee vulnerability rates. Employees must pass a mandatory digital knowledge check module annually with a minimum score of 80% to retain remote network connection access.

Social Media Campaigns

- *Platforms Utilized:* High-engagement public platforms including Instagram, YouTube, LinkedIn, X, and WhatsApp micro-communities to target a broad, 50-million-user demographic.
- *Messaging Themes:* Direct, highly actionable warnings such as "Verify Before You Transfer," "Spot the Deepfake Voice," and "Keep Your OTP Private".
- *Content Types:* Dead-simple 30-second animated reel videos, clear visual infographics charting common financial scam formats, and quick interactive warning quizzes.

Public Awareness Workshops

- *Location:* Distributed across physical civic municipal hubs, local bank branches, rural community centers, and mobile public utility kiosks.
- *Frequency:* Hosted monthly in major smart city sectors and quarterly via mobile cyber-awareness clinics traveling through rural communities.
- *Target Audience:* Senior citizens, small retail shop owners, and digitally illiterate individuals who face elevated risks of automated banking fraud.
- *Topics & Delivery Method:* Hands-on live smartphone configuration demonstrations showing users how to activate two-factor authentication (2FA), set up secure biometric app locks, and file an official complaint via the national cybercrime reporting desk without fear of retaliation.

PART 13: International Cooperation

Define how VIT's will collaborate with other countries and international organizations to combat cybercrime that crosses national borders.

Cybercrime Investigations

The NCSA shall collaborate with international law enforcement agencies to investigate cybercrimes that cross national borders. This cooperation may include sharing digital evidence, forensic reports, technical expertise, and intelligence related to cybercriminal activities. All investigations shall follow established legal procedures and maintain the confidentiality and integrity of evidence.

Information Sharing Agreements

VIT's shall enter into formal information-sharing agreements with foreign governments, international organizations, and cybersecurity agencies to facilitate the secure exchange of cybersecurity information. These agreements shall define data protection standards, confidentiality requirements, information usage limitations, and procedures for responding to cyber emergencies while protecting citizens' privacy and national security interests.

Joint Operations

VIT's shall actively participate in joint international cybercrime operations with foreign governments, INTERPOL, and recognized cybersecurity agencies. When cyberattacks affect multiple countries, the NCSA may establish joint investigation teams to share resources, coordinate evidence collection, identify suspects, and dismantle cybercriminal networks. All joint operations shall follow international law and respect the sovereignty of participating nations.

Extradition Requests

VIT's shall cooperate with foreign governments in extraditing individuals accused of serious cybercrimes such as ransomware attacks, cyber terrorism, large-scale fraud, and data theft. Extradition requests shall be processed through legal channels and approved by competent courts to ensure that the accused receives a fair trial and that human rights protections are maintained throughout the process

Threat Intelligence Exchange

VIT's shall maintain secure threat intelligence sharing agreements with trusted international cybersecurity organizations and partner nations. The NCSA shall exchange information regarding malware campaigns, ransomware groups, emerging vulnerabilities, phishing attacks, and other cyber threats to strengthen national preparedness and enable faster detection and response to cyber incidents

PART 14: Case Studies

Create 3 original fictional cybercrime cases set in VIT's. Each case must be realistic, detailed, and demonstrate how your law would be applied.

Case Study 1: [Give Your Case a Title]

- **Crime Category:** Unauthorized Access / Data Theft.
- **What Happened:** Two days before the high-stakes final mathematics exam, a student worker in the VIT central printing room secretly plugged a thumb drive into the main printing server. They copied the digital master PDF of the exam paper and sold it on a private Telegram channel for ₹2,000 per download. By midnight, hundreds of students had seen the questions, forcing the university to cancel the exam at the last minute.
- **Law Violated:** This act broke Section 66 (Unauthorized Access), Section 66C (Data Theft), and violated Part 6 Corporate Responsibilities because the college left administrative printing systems completely open without logging who accessed them.
- **Investigation Process:** The college DPO flagged the incident to the NCSA within the mandatory 72-hour window. Forensics experts audited the printing server's internal registry logs and matched the exact timestamp of the file export to a specific user credential and a physical USB hardware ID. Campus security cameras confirmed the student's identity at the terminal.
- **Punishment Given:** The student was expelled, sentenced to 3 years in prison, and hit with a criminal fine of ₹5 Lakhs. The university was also penalized under corporate data liability rules with a ₹10 Lakh compliance fine for loose endpoint security.
- **Lessons Learned:** VIT completely banned personal USB storage devices on university networks using automated endpoint protection rules. They also moved all exam printing to a highly secure, air-gapped network zone that requires two separate faculty biometric scans to access or print files.

Case Study 2: [Give Your Case a Title]

- **Crime Category:** Financial Fraud / Phishing.
- **What Happened:** Right in the middle of mid-semester exams, an anonymous user dropped a hyper-realistic AI deepfake video of the College Dean into a student WhatsApp group. The fake Dean claimed the university payment portal had crashed and ordered students to scan an attached Google Pay QR code to clear their dues within 4 hours or be barred from the exam hall. Panicked, 150 students quickly sent ₹5,000 each before anyone realized it was a scam.
- **Law Violated:** Violated Section 66D (Cheating by Impersonation), national Deepfake/Synthetic Media Regulations, and standard financial fraud laws.
- **Investigation Process:** The NCSA Digital Forensics Lab analyzed the video metadata to find the specific AI tool used to generate the clip. Working fast within the "golden hour," tech teams tracked the QR code's digital footprint and coordinated with the bank to freeze the scammer's account, recovering 80% of the students' money.
- **Punishment Given:** The hacker was caught via their phone's location data and telemetry. The court handed down 5 years in prison, a ₹10 Lakh fine, and ordered full repayment to the students.
- **Lessons Learned:** The campus set up an official, verified broadcast channel for all leadership announcements and ran workshops to teach students how to spot edited audio and video markers.

Case Study 3: [Give Your Case a Title]

- **Crime Category:** Ransomware / Cyber Terrorism.
- **What Happened:** During busy morning registration hours at VIT, a student downloaded a malicious file disguised as an "Exam Timetable PDF" from a phishing email while on the open campus Wi-Fi. The ransomware instantly locked up the central student fee database and threw up a screen demand asking each student for an urgent ₹25,000 crypto payment to unlock their files.
- **Law Violated:** Broke Section 66 (Data Theft) and Section 66F (Cyber Terrorism) for crippling a school system during peak hours. The college also failed its Part 6 Corporate Responsibilities by not separating public student Wi-Fi from internal financial networks.
- **Investigation Process:** The IT desk triggered emergency protocols within an hour and snapshot the server's active memory. The campus Data Protection Officer (DPO) reported the attack to the National Cyber Security Authority (NCSA) within the mandatory 72-hour window. NCSA agents traced the network logs, blocked the hacker's external IP addresses, and helped restore the files using the university's offline, air-gapped backups.
- **Punishment Given:** The crypto trail led police straight to a rogue developer syndicate. The main culprit was sentenced to 10 years of rigorous imprisonment and slapped with a ₹50 Lakh fine.
- **Lessons Learned:** VIT killed all unauthenticated open Wi-Fi access, forced mandatory Multi-Factor Authentication (MFA) for all staff, and completely isolated core grading and financial databases onto hidden subnets.

PART 15: Future Recommendations

Based on your research, suggest forward-looking improvements and additions to VIT's cybersecurity law for the next 10 years. Consider how technology will evolve and what new threats will emerge.

AI Security (2025-2035)

VIT's shall establish regulations governing the safe and responsible use of Artificial Intelligence. High-risk AI systems used in healthcare, law enforcement, finance, and critical infrastructure must undergo regular security audits, bias testing, and human oversight reviews. The use of AI for cyberattacks, automated hacking, deepfake fraud, or malicious surveillance shall be treated as a serious criminal offense, while organizations deploying AI systems shall remain accountable for their actions and decisions.

Quantum Computing

To prepare for future threats posed by quantum computing, VIT's shall gradually transition government agencies and critical infrastructure operators to quantum-resistant cryptographic systems. Organizations handling sensitive or classified information must assess their current encryption methods and develop migration plans by 2030. The NCSA shall publish standards and guidelines to ensure a smooth transition to post-quantum security technologies.

Privacy Protection

As technology becomes more advanced, VIT's shall strengthen privacy protections by regulating the collection and use of biometric data, behavioral analytics, location tracking, and smart device information. Organizations shall collect only necessary data, obtain clear user consent, and provide citizens with the right to access, correct, and delete their personal information. Severe penalties shall apply to organizations that misuse or expose citizen data.



Digital Identity

VIT's should establish a secure and privacy-preserving national digital identity framework based on decentralized architecture and strong encryption. Citizens shall have full control over how their identity data is used, and biometric information shall be stored securely with strict access controls. Multi-factor authentication shall be mandatory for all digital identity services to reduce the risks of identity theft and unauthorized access.

Cyber Warfare

VIT's shall recognize cyber warfare as any state-sponsored cyberattack targeting critical infrastructure, government systems, defense networks, financial institutions, or essential public services. The NCSA shall maintain a National Cyber Defense Command capable of detecting, responding to, and mitigating such attacks. In the event of a cyber war, VIT's may coordinate with international allies, strengthen defensive measures, and take lawful countermeasures to protect national security and public safety.

Evaluation Criteria

Your project will be graded based on the following criteria. Review this carefully to maximize your marks.

Criteria		Marks Obtained
Research Quality — depth and accuracy of information, use of real-world examples	20	
Law Structure — completeness, logical organization, and legal clarity of the law	20	
Innovation — originality of solutions, creative approach to cybersecurity challenges	15	
Technical Accuracy — correctness of technical details, cybersecurity concepts, and procedures	15	
Presentation — quality of the PowerPoint presentation and oral delivery	10	
Diagrams & Flowcharts — clarity, accuracy, and professionalism of visual deliverables	10	
Case Studies — realism, detail, and correct application of the law	10	
TOTAL	100	

Deliverables Checklist

Before submitting, ensure you have completed all required deliverables:

Done	Deliverable	Requirements
☐	Project Report	Minimum 25 pages, well formatted, all 15 parts completed, diagrams included
☐	PowerPoint Presentation	15 to 20 slides, covering all major sections of the law
☐	Flowchart: Cybercrime Reporting	Shows complete incident reporting process from detection to resolution
☐	Flowchart: Investigation Process	Shows NCSA investigation steps from complaint to judgment

Done	Deliverable	Requirements
☐	Flowchart: Data Breach Response	Shows data breach response procedure for organizations
☐	Infographic: Citizens' Rights	Visual summary of all citizens' digital rights
☐	Infographic: Cybercrime Categories	Visual taxonomy of all defined cybercrimes with penalties
☐	Infographic: Security Framework	Visual overview of the data protection and corporate security framework

Good Luck!

This project challenges you to think like a lawmaker, a technologist, and a citizen. Research real world cybersecurity laws (India's IT Act, GDPR, Budapest Convention) for inspiration. Be creative, be thorough, and make VIT's digital future secure!